

Closed manifolds with positive Ricci curvature and large nilpotent fundamental group

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Introduction

With E. Bruè and A. Naber we construct a sequence of closed, simply connected Riemannian manifolds (M_k^4, g_k) with

$$\text{Ric} > 0$$

and such that $\text{Isom}(M_k^4, g_k)$ has a subgroup isomorphic to the finite **Heisenberg group**

$$H_3(\mathbb{Z}/k\mathbb{Z}) := \left\{ \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} : a, b, c \in \mathbb{Z}/k\mathbb{Z} \right\} < \text{GL}_3(\mathbb{Z}/k\mathbb{Z}),$$

for every $k \geq 1$.

Outline

Context and motivation

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Section 1

Context and motivation

Fundamental groups and nonnegative Ricci, I

Theorem (J. Cheeger & D. Gromoll '72)

If (M, g) is **closed** with $\text{Ric} \geq 0$, then $\pi_1(M)$ is **virtually abelian**.

Theorem (V. Kapovitch & B. Wilking '11)

If (M^n, g) is closed with $\text{Ric} \geq 0$, then $\pi_1(M)$ has a **nilpotent** subgroup $N < \pi_1(M)$ with **index** $[\pi_1(M) : N] \leq C(n)$.

Remark

Kapovitch, A. Petrunin, and W. Tuschmann obtained the same result in 2005, assuming that $\text{sect} \geq 0$, after earlier progress due to K. Fukaya and T. Yamaguchi in 1992.

In their paper Kapovitch and Wilking asked:

*"can one arrange the subgroup N to have **abelian torsion**?"*

Fundamental groups and nonnegative Ricci, II

In 2018, J. Pan and X. Rong raised:

Conjecture

If (M^n, g) is closed with $\text{Ric} \geq 0$, then $\pi_1(M)$ has an **abelian** subgroup $A < \pi_1(M)$ with **index** $[\pi_1(M) : A] \leq C(n)$.

Remark

Restricting to **closed manifolds** in such conjecture was known to be necessary.

Theorem (G. Wei '88)

*There exists a complete (M^n, g) with $\text{Ric} > 0$ such that $\pi_1(M)$ is **not virtually nilpotent**.*

Fundamental groups and nonnegative sectional

In 1992 Fukaya and Yamaguchi raised:

Conjecture

If (M^n, g) is closed with $\text{sect} \geq 0$, then $\pi_1(M)$ has an **abelian** subgroup $A < \pi_1(M)$ with **index** $[\pi_1(M) : A] \leq C(n)$.

Their conjecture remains open, despite partial progress due to:

- ▶ Rong in 1996;
- ▶ M. Mazur, Rong, and Y. Wang in 2008;
- ▶ Kapovitch, Petrunin, and Tuschmann in 2017.

The Jordan property

Theorem (C. Jordan 1878)

For every $n \in \mathbb{N}$ there exists $C(n)$ s.t. every **finite** subgroup $\Gamma < \mathrm{GL}_n(\mathbb{R})$ has an **abelian** subgroup $A < \Gamma$ with $[\Gamma : A] \leq C(n)$.

Corollary

For every compact **Lie group** $(G, \cdot)$ there is $C > 0$ s.t. every finite $\Gamma < G$ has an abelian subgroup $A < \Gamma$ with $[\Gamma : A] \leq C$.

Remark

If (M^n, g) is compact, then $\mathrm{Isom}(M^n, g)$ is a compact Lie group, by work of **S.-B. Myers** and **N.-E. Steenrod** in 1939.

Question

How much does the constant C obtained from the Corollary depend on g if n is fixed and $\mathrm{Ric}/\mathrm{sect} \geq 0$ on M^n ?

Section 2

Main results

Effective nilpotent actions

Theorem (Bruè-Naber-S. '25)

There exist closed, **simply connected** Riemannian manifolds (M_k^4, g_k) with

$$\text{Ric} > 0$$

such that $\text{Isom}(M_k^4, g_k)$ has a subgroup isomorphic to

$$H_3(\mathbb{Z}/k\mathbb{Z}) := \left\{ \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} : a, b, c \in \mathbb{Z}/k\mathbb{Z} \right\} < \text{GL}_3(\mathbb{Z}/k\mathbb{Z}),$$

for every $k \geq 1$.

Remark

$H_3(\mathbb{Z}/k\mathbb{Z})$ is **nilpotent**.

However, if $A_k < H_3(\mathbb{Z}/k\mathbb{Z})$ are **abelian** subgroups then

$$[H_3(\mathbb{Z}/k\mathbb{Z}) : A_k] \rightarrow \infty \quad \text{as} \quad k \rightarrow \infty.$$

Lifting to free actions

Theorem (Bruè-Naber-S. '25)

*There exist closed Riemannian manifolds (N_k^{10}, h_k) with $\text{Ric} > 0$ such that $\pi_1(N_k)$ is a **degree-two extension** of $H_3(\mathbb{Z}/k\mathbb{Z})$.*

Corollary

There is no constant $C < \infty$ s.t. if (M^{10}, g) is closed with $\text{Ric} > 0$ then there is an abelian subgroup $A < \pi_1(M)$ with $[\pi_1(M) : A] \leq C$.

Remarks on the topology of the examples

Remark

Each manifold M_k^4 is diffeomorphic to the **connected sum** of $(k - 1)$ copies of $S^2 \times S^2$. In particular, it is **spin**.

To build N_k^{10} from M_k^4 we argue as follows:

- ▶ The effective action of $H_3(\mathbb{Z}/k\mathbb{Z})$ on M_k^4 lifts to a **free** action on the **orthonormal frame bundle** of (M_k^4, g_k) .
- ▶ The orthonormal frame bundle of M_k^4 is a $\mathbb{Z}/2\mathbb{Z}$ quotient of the **spin bundle**, which is simply connected.
- ▶ A $\mathbb{Z}/2\mathbb{Z}$ **extension** of $H_3(\mathbb{Z}/k\mathbb{Z})$ acts freely on the spin bundle of M_k^4 . We let N_k^{10} be the **quotient** with respect to this action.

Section 3

Sketch of the construction

Construction of M_k^4 (Part I)

We view $S^3 \subset \mathbb{R}^4 \cong \mathbb{C}^2$ as

$$S^3 := \left\{ (z_1, z_2) \in \mathbb{C}^2 : |z_1|^2 + |z_2|^2 = 1 \right\}.$$

Let

$$\mathcal{B}_k := S^3 \setminus \left\{ (e^{2\pi i \ell/k}, 0), (0, e^{2\pi i \ell'/k}) : 0 \leq \ell, \ell' \leq k-1 \right\}.$$

Remark

The $2k$ removed points belong to S^1 **fibers** corresponding to opposite poles of the **Hopf fibration** $\pi_{\text{Hopf}} : S^3 \rightarrow S^2$.

Remark

There is a natural **action** of $\mathbb{Z}/k\mathbb{Z} \times \mathbb{Z}/k\mathbb{Z}$ on $\mathcal{B}_k$ given by

$$(\ell, \ell') \cdot (z_1, z_2) := (e^{2\pi i \ell/k} z_1, e^{2\pi i \ell'/k} z_2).$$

Construction of M_k^4 (Part II)

Remark

Note that $H_2(\mathcal{B}_k, \mathbb{Z}) \simeq \mathbb{Z}^{2k-1}$ is generated by the **homology classes** of small 2-spheres

$$S_\ell^\alpha = \partial B_r(p_\ell^\alpha), \quad \text{for } \alpha \in \{0, 1\}, \ell \in \{0, \dots, k-1\},$$

$$p_\ell^0 := (e^{2\pi i \ell/k}, 0), \quad p_{\ell'}^1 := (0, e^{2\pi i \ell'/k}).$$

Definition

We define an **integral cohomology class** $[\omega] \in H^2(\mathcal{B}_k, \mathbb{Z})$ by

$$\int_{S_\ell^\alpha} \omega = (-1)^\alpha 2\pi, \quad \alpha = 0, 1, \quad \ell = 0, \dots, k-1.$$

Remark

$[\omega]$ is **invariant** under the $\mathbb{Z}/k\mathbb{Z} \times \mathbb{Z}/k\mathbb{Z}$ action on $\mathcal{B}_k$.

Construction of M_k^4 (part III)

Let $\tilde{\pi} : \mathcal{N}_k \xrightarrow{S^1} \mathcal{B}_k$ be the principal bundle with **Chern class** $[\omega]$.

Remark

By the choice of $[\omega]$, for every $r > 0$ sufficiently small,

$$\tilde{\pi}^{-1}(B_r(p_\ell^\alpha) \setminus \{p_\ell^\alpha\})$$

is diffeomorphic to a **punctured 4-ball**.

$\Rightarrow \mathcal{N}_k$ can be “**completed**” to a closed 4-manifold M_k^4 by adding $2k$ points. The extension is such that:

- ▶ $\tilde{\pi} : \mathcal{N}_k \rightarrow \mathcal{B}_k$ extends smoothly to

$$\pi : M_k^4 \rightarrow S^3;$$

- ▶ the S^1 **action** on $\mathcal{N}_k$ extends smoothly to M_k^4 with the added points as **fixed points**.

Remarks on the topology of M_k^4

Lemma

M_k^4 is simply connected.

Proof.

Since $[\omega]$ is primitive and $\mathcal{B}_k$ is simply connected, $\mathcal{N}_k$ is **simply connected**. Hence, M_k^4 is simply connected as well. $\square$

Lemma

M_k^4 is diffeo. to a **connected sum** of $(k - 1)$ copies of $S^2 \times S^2$.

Proof.

Smooth S^1 actions with finitely many **fixed points** on simply connected 4-manifolds were classified up to **equivariant diffeomorphism** by P.-T. Church and K. Lamotke in 1974. $\square$

Construction of the metric g_k on M_k^4 (part I)

We use a variant of the **Gibbons-Hawking ansatz**.

Let $u : S^3 \rightarrow [-\infty, +\infty]$ be the unique solution of

$$-\Delta_{S^3} u = 2\pi \sum_{\ell=1}^k \left(\delta_{(e^{2\pi i \ell/k}, 0)} - \delta_{(0, e^{2\pi i \ell/k})} \right)$$

odd under the involution $\iota : S^3 \rightarrow S^3$, $\iota(z_1, z_2) := (z_2, z_1)$.

Let $*$ denote the **Hodge star** operator on S^3 w.r.t. the standard metric and set

$$\omega := *du.$$

Lemma

The cohomology class $[\omega]$ represents the Chern class of the principal S^1 bundle $\tilde{\pi} : \mathcal{N}_k \rightarrow \mathcal{B}_k$ in de Rham cohomology.

Construction of the metric g_k on M_k^4 (part II)

- ▶ Consider a **connection** on the principal S^1 -bundle $\tilde{\pi} : \mathcal{N}_k \rightarrow \mathcal{B}_k$ with **curvature form** ω .
- ▶ Let $\theta \in \Omega^1(\mathcal{N}_k)$ be a **principal connection 1-form** corresponding to this connection. In particular, $d\theta = \tilde{\pi}^*\omega$.
- ▶ Consider a smooth, positive and convex potential $V : \mathbb{R} \rightarrow (0, +\infty)$ asymptotic to $|x|$ for $x \rightarrow \pm\infty$.
- ▶ Let $V_\Lambda(x) := \Lambda^{-1} V(x) + 1$, for $\Lambda > 0$.

We introduce a Riemannian metric h_k on $\mathcal{N}_k$ by setting

$$h_k := V_\Lambda(u/k) \tilde{\pi}^* g_{S^3} + \frac{1}{k^2 \Lambda^2 V_\Lambda(u/k)} \theta \otimes \theta.$$

Construction of the metric g_k on M_k^4 (part III)

Lemma

The metric h_k on $\mathcal{N}_k$ **extends smoothly** to a metric $\tilde{g}_k$ on M_k^4 .

Lemma

If Λ is sufficiently large, $\text{Ric} > 0$ away from the $2k$ added points.

Lemma

The metric $\tilde{g}_k$ can be **conformally deformed** to a **positively Ricci curved** metric g_k .

Remark

The conformal factor can be chosen **invariant** with respect to:

- ▶ the S^1 action on M_k^4 ;
- ▶ the $\mathbb{Z}/k\mathbb{Z} \times \mathbb{Z}/k\mathbb{Z}$ action on $S^3 = S^1 \setminus M_k^4$.

Remark

The construction is inspired by many previous appearances of the **Gibbons-Hawking ansatz** in the literature:

[Gibbons-Hawking '78], [LeBrun '92], [Ooguri-Vafa '96],
[Gross-Wilson '00], [Hein-Sun-Viaclovsky-Zhang '22],
[Chen-Viaclovsky-Zhang '20], ... , as well as on the construction
of complete Ricci-flat manifolds with **nilpotent symmetry** on
the **end**, see [Calabi '79], [Tian-Yau '90], [Hein '12],

Remark

Similar uses of **conformal deformations** appeared before
starting from [Aubin '70] and [Ehlich '76].

Construction of the $H_3(\mathbb{Z}/k\mathbb{Z})$ isometric action

Let $K \cong \mathbb{Z}/k\mathbb{Z} \times \mathbb{Z}/k\mathbb{Z}$ be the group of diffeomorphisms of $\mathcal{B}_k$ introduced before.

Let H be the group of diffeomorphisms $\phi : \mathcal{N}_k \rightarrow \mathcal{N}_k$ s.t.:

- (i) ϕ is **equivariant** with respect to the S^1 action on $\mathcal{N}_k$;
- (ii) ϕ is a **lift** of some element $\tilde{\phi} \in K$, i.e., $\tilde{\pi} \circ \phi = \tilde{\phi} \circ \tilde{\pi}$;
- (iii) $\phi^*\theta = \theta$, where θ is the connection 1-form on $\tilde{\pi} : \mathcal{N}_k \rightarrow \mathcal{B}_k$.

Lemma

H acts **isometrically** on (M_k^4, g_k) and it has a subgroup isomorphic to $H_3(\mathbb{Z}/k\mathbb{Z})$.

Main idea of the proof.

$\tilde{\pi} : \mathcal{N}_k \rightarrow \mathcal{B}_k$ restricts to an S^1 bundle of degree k over the **Clifford torus** $\{|z_1| = |z_2|\} \subset \mathcal{B}_k$, whose total space is diffeomorphic to an **Heisenberg nilmanifold** of degree k . □

Thank you for the attention!